

Fachry Dennis Heraldi

Jakarta, Indonesia | +62 882 1185 4646
fd.heraldi@gmail.com
linkedin.com/in/dennisheraldi
dennisheraldi.github.io
github.com/dennisheraldi

Software engineer with a research interest in applied LLMs and natural language processing. Winner of the ICAICTA 2024 Best Paper Award for fine-tuning Llama 2 on sarcasm detection. Currently building LLM-based systems at CSG with prompt engineering, confidence-based decisions, and manual review for safety. Looking to pursue graduate study in AI/ML.

Education

Institut Teknologi Bandung

Bachelor of Engineering in Informatics

August 2020 - October 2024

CGPA: 3.76/4.00 (Cumlaude)

Relevant Coursework: Algorithm and Data Structure, Software Project Management, Database Management, Platform-based Application Development, Parallel and Distributed Systems, Advanced Machine Learning

Publications & Awards

Sarcasm Detection Using Fine-tuned Llama 2

International Conference on Advanced Informatics (ICAICTA 2024)

Singapore

September 2024

Best Paper Award. Fine-tuned **Llama 2** for sarcasm detection on a dataset of **21,000+ samples**, **beating previous benchmarks by 8%** through prompt engineering and parameter-efficient fine-tuning trade-offs that help the model understand nuanced language better.

RubBot, AI-Powered Waste Recognition

Green Our Campus Paper Competition

Institut Teknologi Bandung

November 2023

First Place Winner. Built an AI-based waste recognition system using a **Vision Transformer** integrated with IoT, **improving waste processing efficiency by 90%** to support campus sustainability.

Teaching Experience

Institut Teknologi Bandung

Course and Lab Assistant, Software Engineering Lab

Bandung, Indonesia

August 2022 - July 2024

Course and lab assistant for **Programming Fundamentals** (Haskell, Python), **Software Engineering**, **Software Project Management**, and **Human-Computer Interaction**: ran lab sessions and exams, **mentored 30 junior students through their final projects**, wrote programming problems, and prepared assignment materials together with the course lecturer.

Work Experience

CSG International

Software Developer Engineer II

Jakarta

April 2026 - Present

CSG is a global provider of customer experience, billing, and payment solutions for the telecom industry.

As an early adopter of AI tooling at the company, built an **AI Email Interpreter** in Java with **LangChain** on a self-hosted **Ollama qwen3-8b** model that converts multilingual client emails (**validated in English and Mandarin**) into structured order-status payloads, using **example prompts** and falling back to **manual review** when confidence is low; **delivered in 2 days for a client hackathon** and selected by pre-sales to **demonstrate order-to-inbox capability at MWC Barcelona 2026**.

Won 2nd place at the internal AI competition with an **AI-Assisted API Transformation engine** that converts natural-language instructions into transformation scripts, **automating an API mapping step that previously required manual scripting** during customer onboarding.

Built an **AI Analytics PoC** that exposes the production database through a **chat interface backed by MCP**, giving **business users self-service access to operational data** without writing SQL.

Lead developer for Banking & Financial Solutions (BFS): prepared **10+ international demos at banking events including Sibos**, owned demo environment microservices, and served as **first-line on-call**; also built supporting AI tools (a **PDF reader for financial statements** and an **MCP server that coordinates API calls**) that **cut demo preparation time** for later events.

Software Developer Engineer I

January 2025 - April 2026

Helped the company **win the TForum DTW 2025 award** by building an engine for **intent-driven network API transformations** as the MVP of a Catalyst project, **together with Ericsson**.

Reached zero transaction failures by adding end-to-end recovery with **idempotency, process-engine auto-retry, and stronger internal error handling**, keeping critical business operations reliable.

Paper.id

Software Engineer

Jakarta

November 2023 - December 2024

Paper.id is a platform providing invoicing, payment, and financing solutions for businesses.

Cut message template creation from hours to under a minute by integrating third-party SMS and WhatsApp broadcast services with **concurrent Goroutines processing**.

Built a unified payment engine with a generalized design that handles **different bank behaviors and merchant-specific requirements**, making it easier to integrate with multiple parties while **keeping the system consistent across financial partnerships**.

Reached a 99% transaction success rate by implementing QRIS, direct transfer, virtual account, and cross-border remittance flows that follow **Indonesia's Central Bank standards**, contributing **4 billion IDR TPV** in a month.

Made the system more robust by adding **dynamic table partitioning and automated transaction reconciliation** through RabbitMQ scheduling and async callbacks, **improving query performance by 20%** and making reconciliation reports run instantly.

Suitmedia

Software Engineer Intern

Bandung

May 2023 - September 2023

Built a **Laravel-based CMS** with **35 dynamic menus and 50 API endpoints** from Figma prototypes, with database schemas designed for **smooth integration with the front-end**.

Technical Skills

- Programming Languages:** Java, Golang, PHP, Python, JavaScript, Shell Scripting
- Frameworks & Libraries:** Laravel, React, ViteJS, NodeJS, Fastify, JAX-WS
- Databases:** PostgreSQL, MySQL, ArangoDB, Oracle Database
- Tools & Infrastructure:** RabbitMQ, Kafka, OpenTelemetry, Vault, Consul, Docker, Jenkins, Google Cloud Platform, AWS
- AI / ML:** LangChain, Ollama, OpenRouter, Llama 2, Qwen, Model Context Protocol (MCP), Prompt Engineering, LLM Evaluation